



Extraction of Metals

Contents

- * Extraction of Metals
- * Extraction of Iron from Hematite
- * Extraction of Aluminium from Bauxite



Extraction of Metals

Metals and ores

- The Earth's crust contains metals and metal compounds
- Useful metals are often chemically combined with other substances in rocks called ores
 - A metal ore is a rock that contains enough of the metal to make it worthwhile extracting

Why extraction is a reduction process

- Many metal ores are oxides
- Extracting the metal involves **removing oxygen** from the compound
- This means extraction is a **reduction** process
- Examples of oxide ores include:
 - Iron ore: hematite
 - Aluminium ore: bauxite

Unreactive metals may be found native

- Some very unreactive metals are found as uncombined elements
 - This is because they do not easily react with other substances
- These are called native metals
- Examples include gold and platinum

How the reactivity series links to extraction

- The position of a metal in the reactivity series affects how difficult it is to extract
- The higher a metal is in the reactivity series, the more difficult it is to extract from its ore
- Metals **above carbon** must be extracted using **electrolysis**
 - The most common example of a metal extracted by electrolysis is aluminium
- Metals **below carbon** can be extracted by **heating with carbon or carbon monoxide**
 - A common example of a metal extracted by heating with carbon / carbon monoxide is iron in the blast furnace
- Metals lower in the reactivity series are easier to obtain than metals higher up the series

The extraction method depends on the position of a metal in the reactivity series



Your notes

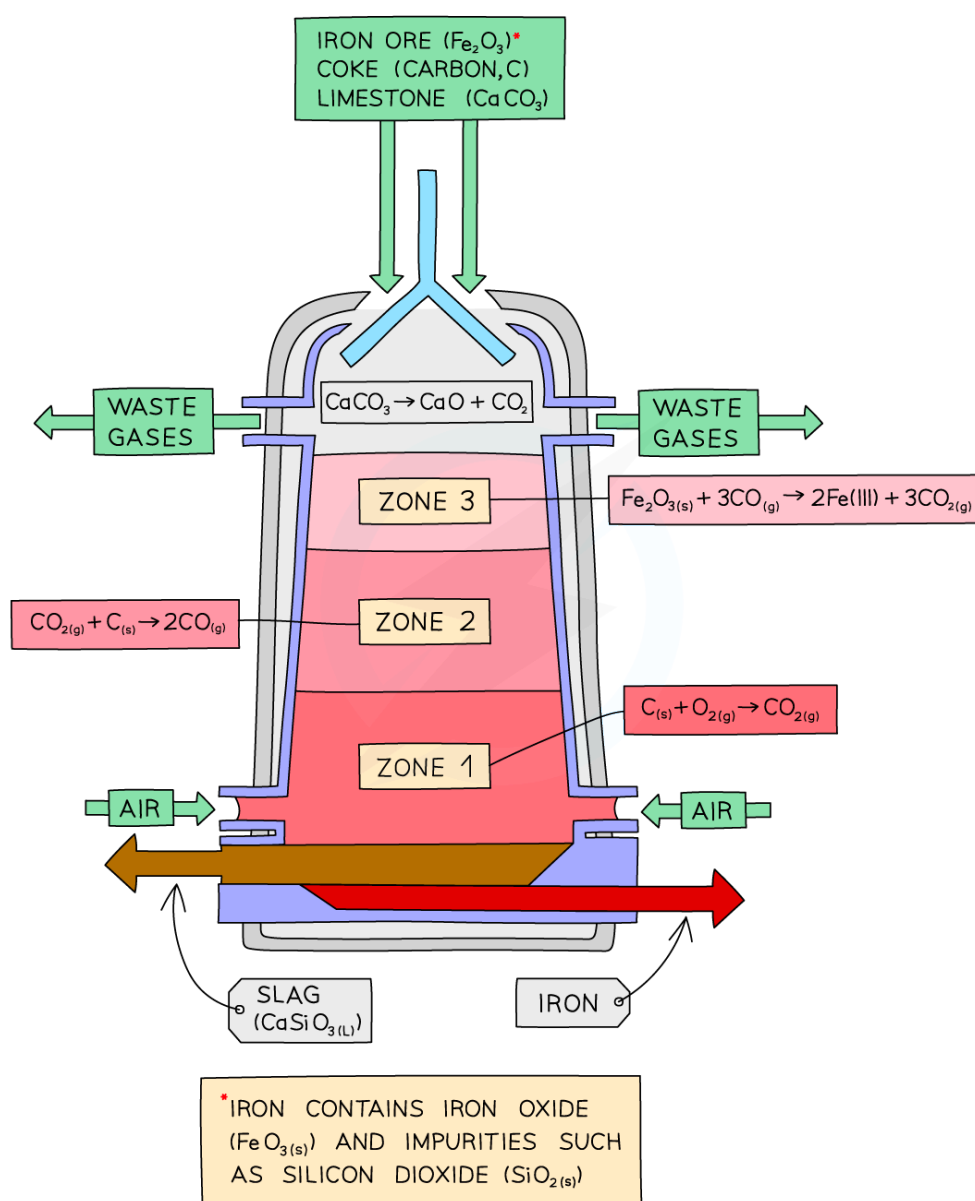
Metal	Method of Extraction
Most reactive	
Potassium	Extracted by electrolysis of molten compounds (chlorides or oxides) Large amount of electricity required so expensive process
Sodium	
Lithium	
Calcium	
Magnesium	
Aluminium	
Carbon	
Zinc	Extracted by heating with a reducing agent such as carbon or carbon monoxide
Iron	
Hydrogen	
Copper	Very unreactive metals Often found as the uncombined element or require little processing to extract
Silver	
Gold	
Least reactive	



Extraction of iron from hematite

- Iron is extracted in a large container called a blast furnace from its ore, hematite
- Modern blast furnaces produce approximately 10,000 tonnes of iron per day

The blast furnace



Copyright © Save My Exams. All Rights Reserved



Diagram showing the carbon extraction of iron

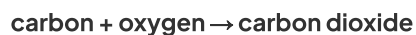
- The raw materials: iron ore (hematite), coke (an impure form of carbon), and limestone are added into the top of the blast furnace
- Hot air is blown into the bottom



Your notes

Zone 1

- Coke **burns** in the hot air to form **carbon dioxide**
- The reaction is exothermic and provides **heat** for the furnace



Zone 2

- At high temperatures, carbon (coke) reacts with carbon dioxide to form carbon monoxide.
- Carbon monoxide is the main reducing agent in the furnace.



Zone 3

- Carbon monoxide **reduces** the iron(III) oxide to iron
 - Iron is extracted by reduction because oxygen is removed from iron(III) oxide
- The molten iron collects at the bottom of the furnace and is tapped off



- Limestone (calcium carbonate) is added to the furnace to remove impurities in the ore
 - The calcium carbonate in the limestone **thermally decomposes** to form calcium oxide



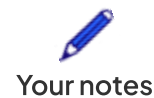
- The calcium oxide formed reacts with the silicon dioxide, which is an impurity in the iron ore, to form calcium silicate
- This melts and collects as a molten **slag** floating on top of the molten iron, which is tapped off separately



Examiner Tips and Tricks

For Core students, the symbol equations are **not** needed for the different reactions involved in the extraction of iron from hematite.

Equations for extraction of iron from hematite



Extended tier only

Zone 1

- The burning of carbon (coke) to provide heat and produce carbon dioxide:



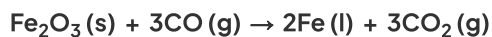
Zone 2

- The reduction of carbon dioxide to carbon monoxide:



Zone 3

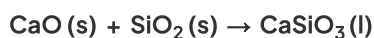
- The reduction of iron(III) oxide by carbon monoxide:



- The thermal decomposition of calcium carbonate (limestone) to produce calcium oxide:



- The formation of slag:

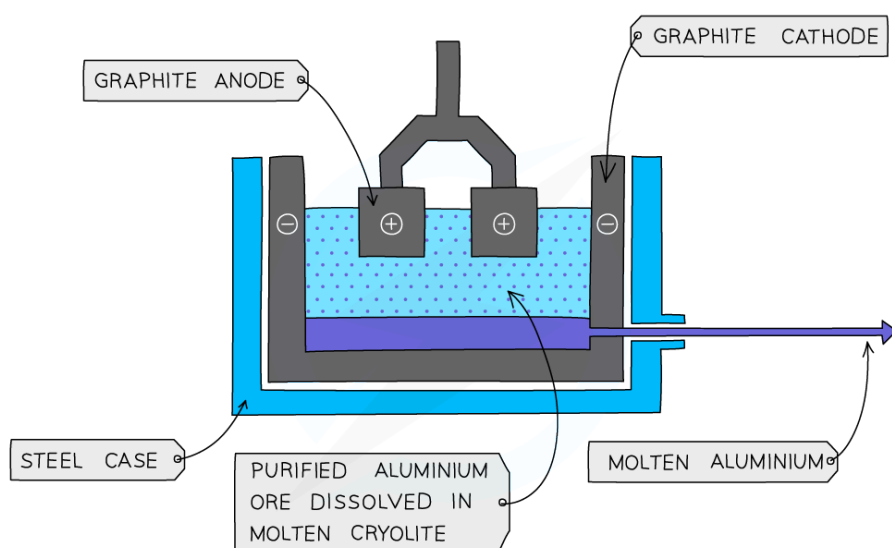




Extraction of aluminium from bauxite

- Aluminium is a reactive metal, above carbon in the reactivity series
- Its main ore, is **bauxite**, which contains aluminium oxide
- Aluminium is higher in the reactivity series than carbon, so it cannot be extracted by reduction using carbon
- Instead, aluminium is extracted by **electrolysis**

The electrolysis of aluminium



Copyright © Save My Exams. All Rights Reserved



Diagram showing the extraction of aluminium by electrolysis



Examiner Tips and Tricks

If you are a Core student, you do **not** need to explain the process of extraction of aluminium by electrolysis.

The process of aluminium extraction by electrolysis

Extended tier only

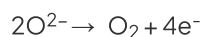


Your notes

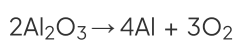
- Bauxite is first **purified** to produce aluminium oxide, Al_2O_3
- Aluminium oxide is then dissolved in **molten cryolite**
 - This is because aluminium oxide has a melting point of over 2000°C which would use a lot of energy and be very expensive
 - The resulting mixture has a lower melting point without interfering with the reaction
- The mixture is placed in an electrolysis cell, made from steel, lined with graphite
- The **graphite** lining acts as the negative electrode, with several large graphite blocks as the positive electrodes
- At the **cathode** (negative electrode):
 - Aluminium ions gain electrons (reduction)
 - Molten aluminium forms at the bottom of the cell
 - The molten aluminium is siphoned off from time to time and fresh aluminium oxide is added to the cell



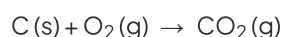
- At the **anode** (positive electrode):
 - Oxide ions lose electrons (oxidation)
 - Oxygen is produced at the anode:



- The overall equation for the reaction is:



- The carbon in the **graphite** anodes reacts with the oxygen produced to produce CO_2

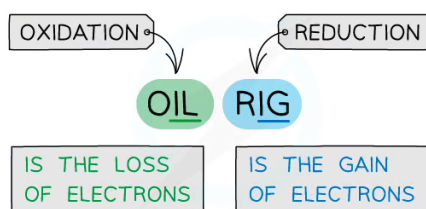


- As a result the anode wears away and has to be replaced regularly
- A lot of **electricity** is required for this process of extraction, this is a major **expense**



Examiner Tips and Tricks

Use OIL RIG to remember whether oxidation or reduction has occurred at the electrodes:



Your notes